Abstract We describe Beacon, a bidirectional encoder pretrained with a masked token objective. Unlike left-to-right language models, Beacon conditions on context from both directions at every layer, which produces representations that transfer well to sentence level tasks.

1 Pretraining objective We mask fifteen percent of the input tokens at random and train the model to recover them. Because the masked token is never revealed to the encoder, the model must rely on surrounding context alone. A second objective predicts whether two segments were adjacent in the source corpus.

2 Fine-tuning For each downstream task we add a single output layer and fine-tune every parameter end to end. Fine-tuning is inexpensive relative to pretraining: most tasks converge within three epochs on a single GPU.